

Andy Banh

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EDUCATION

York University

Hons. Bachelor of Science in Computer Science

Toronto, Ontario

Expected April 2028

Relevant Courses

Advanced OOP, Data Structures and Algorithms, Discrete Mathematics, Theory of Computation

EXPERIENCE

Sole Proprietorship

April 2025 - Present

- Founded and managed a **sole proprietorship**, managing full operations including inventory, sales, and customer service, for over **500+** products
- Analyzed sales data and trends to optimize product sourcing and pricing, generating over **\$60k** in revenue in 2025

PROJECTS

Form Coach | *Next.js, Spring Boot, TypeScript, Python, OpenCV, Mediapipe, FastAPI, Supabase*

- Designed and developed a full-stack computer vision platform utilizing **Next.js** and **Spring Boot** and **FastAPI**, enabling real time pose-detection and movement analysis using **OpenCV** and **Mediapipe**
- Implemented real time bi-directional **WebSocket** communication following **STOMP Protocols** to stream frames and deliver instant feedback
- Processed body landmarks using **OpenCV** and **Mediapipe** and used **NumPy** to calculate joint angles and automate feedback
- Optimized frame processing pipeline to achieve real time analytics at **10+ FPS** with sub-second latency

Bathroom Buddy | *React.js, Spring Boot, Tailwind CSS, PostgreSQL, Supabase*

- Developed an enterprise-style full-stack supply request management system for submitting and tracking facility supply requests using **PostgreSQL**, **Spring Boot**, and **React.js**, enhancing customer experience
- Utilized **Bucket4j** Java library to implement API rate limiting using Bucket4j token-bucket algorithm to restrict abusive traffic, improving stability during high traffic requests
- Secured **REST API** using **Spring Security Architecture** and **JWT authentication** to secure protected endpoints and ensure secure user data
- Increased data-transfer efficiency and safety by structuring backend operations around **DTO** patterns

Discrete Math Chatbot | *Next.js, TypeScript, Python, LangChain, Chroma, FastAPI*

- Built an AI-powered study assistant with **RAG** agents powered by **OpenAI** as LLM and **Fast API**, to provide users with accurate explanations of Discrete Math Topics
- Implemented **Retrieval-Augmented Generation** (RAG) pipeline with **LangChain** and **ChromaDB**, storing data in vectors allowing for context-aware answers
- Designed a responsive frontend in Next.js and TypeScript using Tailwind CSS and DaisyUI, providing sleek and seamless user experience

Expense Tracker | *React.js, Node.js, Express, Tailwind CSS, MongoDB*

- Built a **M.E.R.N full-stack web application** using Tailwind CSS, and React.js to help users track and manage personal expenses
- Improved security by implementing an authentication page allowing for users to sign up and login, using Bcrypt to hash passwords to be securely stored in MongoDB
- Utilized **Postman** to conduct testing on API endpoints, ensuring appropriate JSON responses

TECHNICAL SKILLS

Languages: Java, JavaScript, TypeScript, C, Python, Bash, SQL, HTML, CSS

Frameworks & Libraries: Spring Boot, React.js, Next.js, FastAPI, Tailwind CSS, JUnit

Databases: PostgreSQL, MongoDB, Supabase

Developer Tools: Git, Docker, AWS, Linux, Maven, Postman, IntelliJ, Eclipse